Predicting Port Congestion and Shipment Delays

Technical report for the Aiqon take-home technical challenge
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Summary

- Predicts, 48 hours before a ship's scheduled arrival, the chance that the port call is delayed by over 12 hours.
- Returns a probability, a confidence score (0–100), the main causes and a suggested action.
- Trained and tested only on a simulated year at five ports, where delays come from ships queueing for berths.
- ROC-AUC (Area Under the Receiver Operating Characteristic curve) of 0.85, against 0.81 for a rule of thumb.

1. Problem and label

- User: a planner deciding two days ahead whether to re-time the arrival, switch port or warn the customer.
- Label: congested if anchorage wait + berth overrun is more than 12 hours (Figure 1).
- Inputs must be known 48 h before the ETA (Estimated Time of Arrival); a unit test guards against leakage.

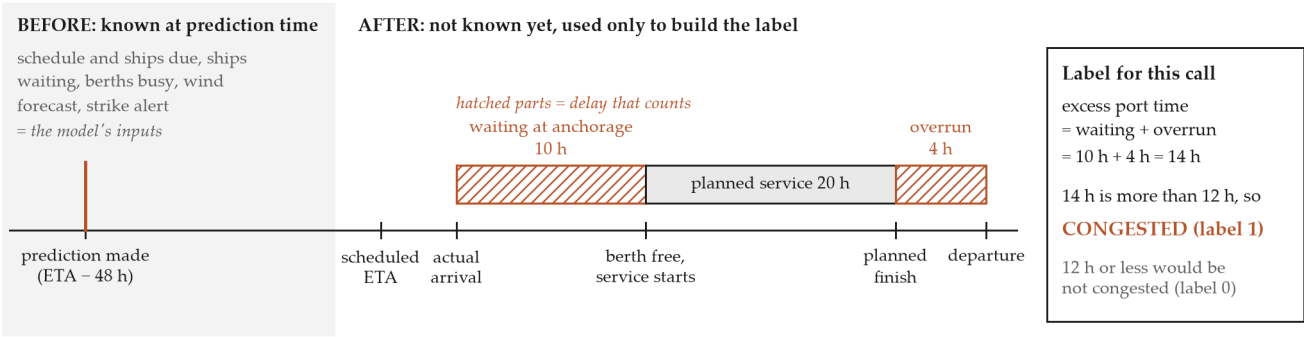


Figure 1: How one port call is labelled. Only the left part is available to the model.

2. Data

- Simulated: building port calls from raw AIS (Automatic Identification System) data would not fit in three days.
- 5 ports (3–12 berths), one year, hour by hour; ships are served first come, first served.
- Seasonal peaks, schedule surges, storms, strikes and night shifts slow the work.
- 10,723 calls simulated, 500 sampled (23% congested).
- Inputs are noisy on purpose: a wind forecast instead of real wind, imperfect strike alerts, 5% missing values.

3. Features

- All are available two days ahead (Table 1).
- Key feature: pressure index = (ships waiting + berths busy + ships due before ours) ÷ berths.
- Missing values are filled with the training median and lower the confidence; invalid inputs are rejected.

Table 1: Model inputs (TEU = twenty-foot equivalent unit, a measure of ship size).

Group	Features	Source in practice
Traffic	ships waiting, berths busy, 7-day utilisation, arrivals due, pressure index	AIS, terminal systems
Port	number of berths, port, share of calls delayed in the last 30 days	port statistics
Weather, labour	wind forecast, storm forecast, strike alert	forecasts, port authority
Vessel, time	ship size (TEU), hour, weekend, month	schedule

4. Model and outputs

- Split by time: 350 calls to train (Jan–Sep), 75 to calibrate, 75 to test (Nov–Dec).
- Compared a rule of thumb, logistic regression and LightGBM (Light Gradient-Boosting Machine).
- Chosen: logistic regression, which beat LightGBM in all 5 time-based folds (ROC-AUC 0.86 vs 0.81).
- Domain rule: more traffic, wind, strikes or ship size can never lower the risk.
- Confidence = 50% agreement of 20 resampled models + 25% data completeness + 25% similarity to past data.
- Causes = each input's contribution (its SHAP, SHapley Additive exPlanations, value), grouped into factors.
- Act when risk ≥ 0.17, assuming a missed delay costs 5 times a false alarm.

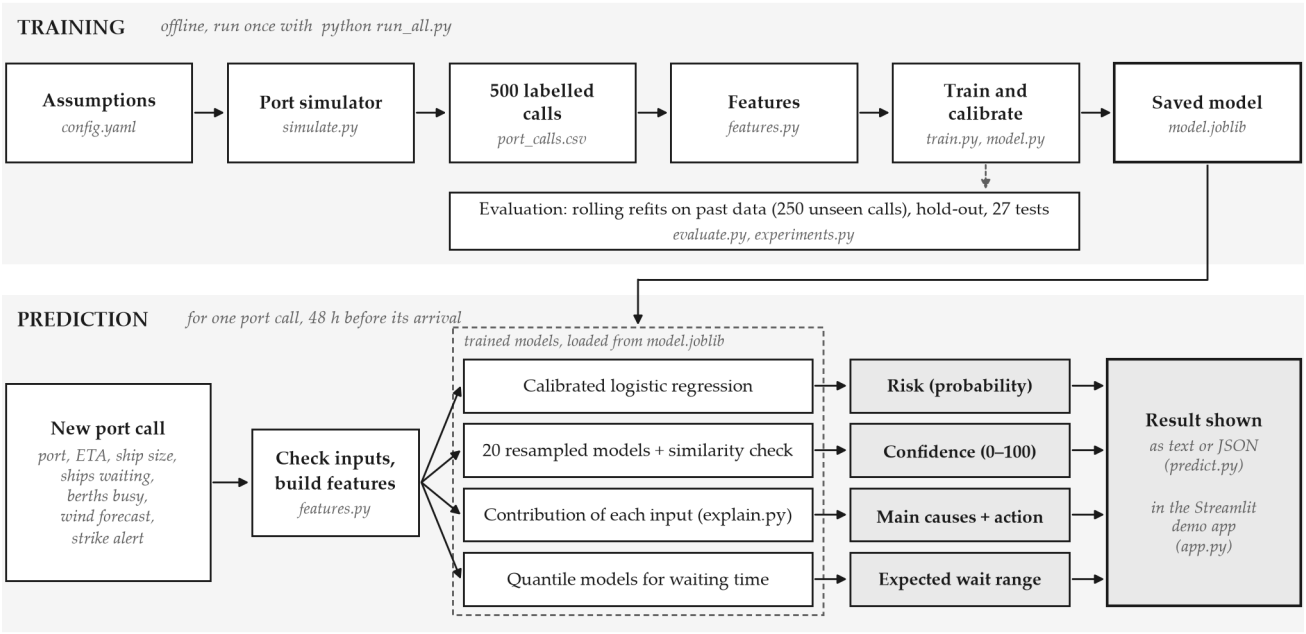


Figure 2: System architecture. The training pipeline runs once; the prediction path runs for each port call.

Example output for one call (shortened):

```
HIGH congestion risk (>99%, confidence 63%). Expected anchorage wait ~38h (80% range 9-48h).
  + +6.70  heavy traffic: 22 vessels due in the 48h before ETA, 6 waiting at anchorage, 6/6 berths occupied
  + +1.85  storm risk: forecast wind 35 kn (cranes stop above 32 kn)
  * Request a later berth window or slow-steam to arrive after the peak
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5. Results

- Main test (rolling origin): refit 5 times on past data, test on the next period; 250 unseen calls in total.

Table 2: Results on the 250 rolling-origin calls; last column is the 75-call hold-out. CI = confidence interval. PR-AUC = area under the precision–recall curve. Brier = error of the probabilities (lower is better).

Model	ROC-AUC [95% CI]	PR-AUC	Brier	Precision / recall at 0.17	Hold-out ROC-AUC
Logistic regression	**0.85** [0.79–0.90]	**0.76**	**0.13**	0.48 / 0.88	**0.87**
LightGBM	0.81 [0.74–0.86]	0.72	0.15	0.45 / 0.83	0.82
Rule of thumb	0.81 [0.75–0.86]	0.62	–	0.64 / 0.60	0.83

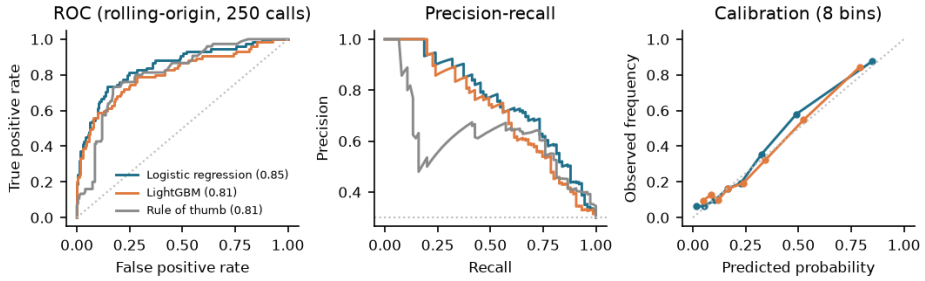


Figure 3: ROC curve, precision–recall curve and calibration on the 250 rolling-origin calls.

- Catches 88% of congested calls; decision cost 236 against 350 for the rule of thumb.
- Confidence is meaningful: the Brier score falls from 0.18 (least confident third) to 0.08 (most confident).
- Stable: ROC-AUC 0.81 ± 0.05 on 5 other simulated years, and 0.88 with 15% missing data.
- Weakness: storms at quiet ports (2 of 8 storm-hit calls missed).

6. Limitations and next steps

- Simulated data only, with simplified operations (no priorities, tides or diversions).
- The domain rule is an assumption; the hold-out is small (75 calls); the costs are placeholders.
- Next: a real AIS data pipeline, per-port retraining, a storm risk floor, and an API (Application Programming Interface) for the JSON (JavaScript Object Notation) output.